RISC-V LOCAL INFERENCE

Qwen3.5 Inference on the SpaceMiT K1

Infrastructure Optimization and Performance Evaluation

Evidence cutoff: October 12, 2026 (Asia/Singapore)

Platform	MUSE-Pi-Pro, K1/X60, 16 GiB RAM
Models	Qwen3.5 2B and 4B
Implementation	SpaceMiT llama.cpp fork
Package	Technical report and evidence appendix

This report separates verified release results, later candidates, and prediction limits. Its measurements do not establish a cumulative project speedup.

github.com/MoyaMryia/riscv-accl

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Project report - evidence cutoff: October 12, 2026, Asia/Singapore
Platform: MUSE-Pi-Pro, SpaceMiT K1/X60, 16 GiB RAM
Models: Qwen3.5 2B and 4B conversational checkpoints
Implementation: SpaceMiT's open-source llama.cpp fork

Abstract

This project implements and tests local Qwen3.5 inference on the MUSE-Pi-Pro with the SpaceMiT K1/X60 and 16 GiB RAM. The main release retains Q4_0 weights, F16 K/V, complete inputs, and direct decoding. The 256-bit RVV attention path reduces 8,192-token first-token latency by 49.2% on 2B and 64.0% on 4B. Separate IME1 routing tests reduce 64-token decode time by 14.40% and 28.62%. These comparisons use different controls and do not establish a cumulative speedup.

Later candidates improve selected prefill or decode workloads on the optimized baseline. Their complete-answer tests contain 48 identical pairs, but shared model errors remain. The three-candidate combination fails one decode gate with a 3.0220477% regression against the attention-only candidate. Its second round and complete-answer stages do not run. The verified October 7 release remains the pinned package; routing stays opt-in.

The project also provides a portable performance evaluator and native operator data. V7 covers all nine reserved groups with wide intervals, but its point directions agree with measurements in only 9/27 cases. The first operator batch contains 1,344 observations across 32 configurations. These training data do not establish independent prediction accuracy. K1 remote access fails before a confirmed second batch. Recovery remains pending; this document does not authorize new board work.

Report status and evidence levels

Evidence level	Meaning in this report
Implemented	Source code or an executable tool exists. This status does not establish a performance gain.
Verified	The stated checks pass for the named configuration and records.
Failed gate	A measured result does not meet its declared criterion. Later versions do not remove that result.
Pending	A measurement or independent check is incomplete.
Not implemented	The project does not provide the stated capability.

This report separates the verified release, later candidates, and performance-model work. Dated reports describe their original campaigns. The evidence appendix maps claims to records and tools. The report and appendix form one review package. No external competition submission is recorded.

1. Objective and scope

The task deploys both required Qwen3.5 models locally through the SpaceMiT fork. It targets prefill, decode, and long-context inference. Deliverables include baseline comparisons, executable code, and reproducible test results.

The current optimization mission concerns infrastructure. It retains complete inputs, Q4_0 weights and F16 K/V. Retrieval, prompt shortening, token pruning and model replacement are excluded. Earlier Q8_0 and speculative experiments are included as supporting evidence with their separate settings and limits. All model generation runs locally. An optional cloud judge is used only for evaluation of completed answers, including cache reuse and MTP comparisons; inference does not require that judge.

Requirement coverage

Requirement	Evidence and boundary
Both 2B and 4B models	Both have local inference, matched prefill comparisons and routing decode comparisons.
SpaceMiT open-source llama.cpp	Current release pins official-fork `a990751` and measured source hashes. The historical patch stack targets `5ad05d8`.
Baseline and optimized results	Matched RVV off/on and GEMM staging/bypass tables below; each comparison states its baseline.
Quantifiable infrastructure improvement	RVV reduces long-prompt TTFT; the new routing screen improves both phases on both models.
Deployment, scripts and tests	Integration guide, patch helper, native harnesses, tmux runners and archived evidence are provided.
Generation above 1 token/s	Demonstrated in short/moderate-context decode tests; this is not guaranteed at every long context.
Cross-architecture comparison	Excluded from the release requirements at the user's direction.

The original requested checklist and competition background have different scopes. This report states the measured coverage rather than treating every suggested advanced challenge as completed.

2. Platform, models and measurement method

The board runs Bianbu 2.3.5 with a RISC-V Linux environment. The evaluated backend uses RVV and SpaceMiT IME1; four inference workers are used because the IME-capable cores are 0–3. Build instructions specify GCC 14 and the matching SPERT runtime. The latest runtime audit records the performance governor at 1.6 GHz. Runtime libraries and original build flags are preserved in experiment provenance.

The principal benchmark files are:

- `Qwen3.5-2B-MTP-Q4_0-embQ4_0-dv64k.gguf`
- `Qwen3.5-4B-MTP-Q4_0-embQ4_0-dv64k.gguf`

These are the project's conversational checkpoint derivatives; their filenames omit the word `Instruct`. MTP tensors are present, but the principal RVV and routing results use **direct decoding**. Weight and cache precision are distinct: Q4_0 weights with F16 K/V is the common current configuration.

Controls and metrics

Item	Protocol
Model/runtime controls	Same model and full prompt within each comparison; four workers, batch/microbatch 32, flash attention; direct greedy decoding.
Run order	Four cold ABBA requests for each routing comparison: control, candidate, candidate, control. Earlier campaign repetitions are stated separately.
Cache isolation	Cold requests require zero reused prompt tokens and the complete intended input count.
TTFT	Client time to first token; includes work visible to the requesting client.
Prefill time	Server prompt-processing time; a one-token request isolates this phase but does not establish answer quality.
Decode rate/time	Server generation measurements after prefill; latest screen captures all 64 requested tokens.
Memory	Whole-process RSS is distinguished from estimated KV payload.

Item	Protocol
Correctness	Native finite-output/guard checks plus input, token and text hashes, completion counts and actual kernel activation logs.

For the routing screen, a candidate's mean time reduction must exceed both 3% and the observed full control range. Required comparisons must have no clear regression. This is a predeclared engineering gate, not a statistical significance test. Two samples per arm do not establish a population confidence interval. Exact output agreement is limited to the tested inputs and settings.

Time reduction and throughput increase use different denominators. For control time t_b and candidate time t_c, the time reduction is

$$R_t = 100 \left(1 - \frac{t_c}{t_b}\right) \%.$$

For equal output counts, the throughput increase is

$$G_q = 100 \left(\frac{t_b}{t_c} - 1\right) \%.$$

The tables state whether they use means, medians, or median paired ratios. A ratio of medians need not equal the median paired ratio.

3. Implemented optimizations

3.1 RVV attention for the K1's vector width

The original wide-vector prototype did not activate on the K1. The subsequent 256-bit implementation enables the F16-KV wide-head prefill path and is packaged as optional patch 0009. It is enabled with `SPINE_FA_WIDE_TILE=1`; activation is checked in the server logs. It improves prompt processing and does not establish a single-token decode gain. Quantized K/V and other model head shapes are outside its demonstrated dispatch.

3.2 Production Q4_0 GEMM routing

Production attribution identified substantial staging-copy work: 9.55% / 9.93% of summed instrumented prefill worker elapsed time for 2B / 4B. These sums include overlapping waits and instrumentation overhead, so they are not removable wall-time estimates. The new code instead tests the existing direct-memory branch in production `forward_mul_mat`.

Eligibility is restricted to Q4_0 K32/N16 traits using IME1, with K divisible by 32 and N divisible by 16. The selected buffer is set to null with size zero; weight packing, activation quantization and IME arithmetic are retained. The generated library is isolated from the original build.

`SPINE_K1_GEMM_ROUTE`	Selection	Validation status
Unset, invalid or 0	Original staging	Control
1	Bypass for $M > 1$	Numerical, operator and model prefill gates passed
2	Bypass for $M = 1$	Numerical, operator and 64-token model decode gates passed
3	Bypass for both	Combined 8k phase gates pass on both models; overall quality gate remains open

Mode 2 can affect single-row work during prefill as well as decoding. The current TCM API reports unavailable/fake/zero geometry, while SPERT supplies non-null 128 KiB worker buffers. Their physical placement is unverified; the measured result concerns staging policy under this runtime.

3.3 Earlier kernel and runtime work

The packaged work also includes recurrent-state in-place write-back, microbatch selection, Q8_0 IME1 support, Q4_0 M4 scale construction and workload-specific speculative policies. Q8_0 IME1 raised 4B pp64 from 1.117 to 7.472 token/s and tg32 from 0.866 to 1.349 token/s in its separate higher-precision comparison. Q4_0 remained faster overall.

M4 scale construction raised 4B Q4_0 pp128 from 9.058 to 9.360 token/s (3.34%). Corrected analysis gives a conditional independent-launch Welch 95% interval of +0.095 to +0.511 token/s, with only two launches per arm. The roughly 1.8% English/Chinese mapped-MTP gains have intervals including zero. These small effects remain workload-specific. Detailed implementation, attribution and statistics are linked in the evidence appendix.

4. Matched performance results

4.1 RVV attention versus the generic attention path

Both arms use the same model and configuration within each comparison, Q4_0 weights, F16 K/V, direct decoding and 32 generated tokens. The baseline here has RVV wide attention disabled.

Model / input tokens	Generic TTFT	RVV TTFT	TTFT reduction	Launches per arm
2B / 2,048	113.78 s	92.45 s	18.7%	2
4B / 2,048	343.95 s	231.54 s	32.7%	2
2B / 8,192	862.62 s	438.05 s	49.2%	2
4B / 8,192	3,194.99 s	1,150.42 s	64.0%	2
2B / 16,384	2,808.28 s	1,124.57 s	60.0%	1
4B / 16,384	11,062.98 s	3,216.23 s	70.9%	1

All these comparisons have complete matching token/text hashes. The 2k/8k rows use the isolated build; 16k uses the integrated build. The single-pair 16k observations need replication. Decode and peak RSS were effectively unchanged within these attention comparisons.

A single RVV-enabled 2B request with an actual 32,768-token prompt completed at 3,291.21 s TTFT, 9.96 prefill token/s and 2,768.4 MiB peak RSS. Its 32 output tokens completed; decode was 0.95 token/s. This establishes bounded feasibility, with no matched 32k control or answer-quality claim. 4B/32k and 64k are unmeasured.

4.2 New GEMM routing versus the current optimized baseline

The baseline already includes accepted wide attention and layout 0. Each row has two launches per arm. Prefill rows use mode 1 and one output token; decode rows use mode 2, a full 256-token prompt and 64 captured output tokens.

Model / measured phase	Control mean	Candidate mean	Time reduction
2B / 512-token prefill	22.052917 s	20.625322 s	6.47%
4B / 512-token prefill	56.740043 s	52.304415 s	7.82%
2B / 2,048-token prefill	92.634959 s	86.709042 s	**6.40%**
4B / 1,024-token prefill	115.551047 s	106.596561 s	**7.75%**
2B / 64-token decode	17.016656 s	14.566911 s	**14.40%**
4B / 64-token decode	43.553046 s	31.087887 s	**28.62%**

Decode throughput rose from **3.76 to 4.39 token/s** for 2B and **1.47 to 2.06 token/s** for 4B. Mode 2 showed no clear prefill regression. All six comparisons cleared their declared gates. These are time reductions; throughput increases have a different percentage denominator.

The board run completed successfully in 38.16 minutes. Validation included 104 real production graph cases per arm across five arms, 144 production operator records, and 24 model requests. Every model comparison matched input/output hashes and uncached counts. All 236 collected artifact hashes were independently verified, and end-of-run checks matched 46 original source/object/build files. The verifier recomputes the required gates from individual records.

These initial routing modes were tested separately. Their gains must not be added or combined with older RVV percentages to claim cumulative acceleration. The subsequent staged validation completed in 4.16 hours with 83 artifact hashes verified. Combined mode 3 at 8192 input/64 output tokens reduces prefill/decode time by 4.43%/9.49% on 2B and 5.79%/15.06% on 4B. All eight requests match hashes and cold counts. The 12 full-document answers stop naturally and match paired baseline text. Both 2B arms lack required tracing citations, so the overall gate remains open. The protocol skips 4B/32k. These are bounded engineering pilots.

4.3 Independent memory-bandwidth and routing confirmation

The October 6 campaign completed 150 native full-read scans, 24 cold direct-inference requests and two separate prefill profiles. The four-core model-sized streaming reference is approximately 7 GB/s. Main weight-read proxies, excluding MTP draft tensors, are 1.060 GB for 2B and 2.369 GB for 4B, giving weight-only references of 6.60/2.94 tok/s. These are traffic references rather than measured DDR utilization or attainable inference guarantees.

At 256 prompt tokens, route-0/route-3 decode medians are 3.79/4.48 tok/s for 2B and 1.48/2.06 for 4B; median paired throughput gains are 18.01%/38.70%. At 2,048 tokens the medians are 3.31/3.83 and 1.22/1.62, with paired gains 15.47%/32.36%. All requests generate the same 64 capped output tokens and match their paired control. This confirms the existing routing mechanism; these gains must not be multiplied by previous routing measurements.

The profiles identify convolution and synchronization as investigation targets, but sampled prefill CPU shares are not wall-time savings or decode attribution. This campaign adds no useful-answer qualification and does not demonstrate that the hardware limit has been reached.

5. Later candidates and the stopped combination

5.1 Independent candidates on the optimized baseline

These comparisons retain the fixed 2B/4B GGUF files, four workers, batch and microbatch 32, F16 K/V, route 3, temperature 0, and seed 42. MTP is disabled. Prefill tests use 512 or 2,048 input tokens. The ranges below describe observed configurations across two rounds. They are not population confidence intervals.

Candidate	Change	Measured model result	Scope
C1	Hybrid convolution; original path for small batches	One-output-token TTFT falls 4.26–6.53%	64 cold requests across two rounds; 24 complete answers
C2-r2	GDN register kernel reduces intermediate state access	One-output-token TTFT falls 4.35–5.95%	96 cold requests across the screen and two rounds; 24 complete answers
C3	C1 with the older C2	TTFT falls 8.35–9.73% against its disabled control; 3.53–4.63% against C1	128 cold requests across two controls and two rounds; 24 complete answers

Candidate	Change	Measured model result	Scope
B1-r2	Shared attention path with a fallback for small work groups	At 2,048 input/65 output tokens, decode time falls 10.48–10.64% on 2B and 16.95–20.32% on 4B	64 cold requests across two rounds; 24 complete answers

GDN means gated delta network, the recurrent-state component in these models. C3 contains the older C2, not C2-r2. All four versions complete 96 answers with 48 identical text pairs. Shared arithmetic, citation, code, and Chinese-language errors remain. These tests do not show an answer-quality improvement.

C2-r2 increases 4B/512/65 decode time by 2.9480% in its first round. This result is near the 3% regression gate and is not a decode gain. The old controller deletes some native dumps before the later review. Saved logs cannot replace a fresh comparison of those dumps. Complete-answer requests also lack full operating-system thread snapshots. The candidate review records these limits.

5.2 Combined-r1: a valid performance stop

The combination uses one library with independent switches. C0 disables the three new candidates but retains wide RVV attention and route 3. B enables B1-r2 only. P enables C1 and C2-r2. F enables B1-r2, C1, and C2-r2. C0 is an optimized control, not the initial vendor source.

The run stops on October 11, 2026, at 18:28 Asia/Singapore with exit code 0 and state `performance-gate-stopped`. The October 12 audit recomputes all 48 comparisons from the first round's 96 requests. It confirms zero reused prompt tokens, output receipts, input hashes, and saved thread snapshots. One required comparison fails: F versus B for 4B/2,048 input/65 output tokens. Decode time rises 3.0220477%, above the declared 3% gate. The full control range is 0.0726418%.

Each table value is the mean of two requests in that arm. All times are milliseconds.

Model / input tokens	B prefill	F prefill	B decode	F decode
2B / 512	20959.891	18769.821	14913.766	14920.559
2B / 2048	88041.229	79032.342	15716.909	15714.604
4B / 512	52863.175	47537.975	30820.230	31336.530
4B / 2048	226049.395	203459.167	33608.751	34624.424

For the failed configuration, F prefill time falls 9.9935% against B. F decode time also falls 17.5476% against C0. Neither result clears the required F-versus-B regression gate. The second round and complete-answer tests do not run. The combination does not complete acceptance.

The audit does not recollect or compare all 536 formal state dumps. It checks saved input and text hashes, not reconstructed inputs or new answers. Startup thread snapshots allow CPU 0–7 and do not establish worker positions during decode. This gap does not prove incorrect core placement. Both failed-comparison arms emit 2,048 shared-attention log lines per request. The evidence does not establish extra attention logs as the cause. Logging cost remains part of the measured result.

5.3 Local revision and comparison boundary

The local dispatch revision changes one GDN condition order. It returns single-token and non-128-state calls before it reads the prefill switch. All 90 dispatch combinations select the same paths. The cross-compiler syntax check and four local tool checks pass. Full numerical checks, target assembly, operator tests, and model tests remain pending. This revision is not a verified fix for the regression. The stopped combination remains a failed result.

The original/current comparison contains historical phase measurements and explicit version identities. Its October 11 pending entries predate the stop audit. Use the October 12 audit for the combination's current status. No matched test compares the initial vendor source directly with a final accepted combination. The separate RVV, routing, and candidate gains must not be added or multiplied.

6. Negative results and correctness limits

Candidate	Measured outcome	Decision
Compact attention layouts	Original 2k pilot gains of 0.89% / 1.98%; later fast-screen model gains below 3%	Retain layout 0
Further IME scheduling / scale gathering	Numerical checks pass; operator screens do not qualify	Retain original schedule
Hybrid SSM convolution/RVV	Exact graph/state checks and 24 natural answers pass; total latency falls 2.92% / 2.13%, below the adoption gate	Disabled in the release
Shared dense FFN activation packing	Duplicate work is only 0.024–0.035% of summed GEMM worker elapsed in the decode audit	Implementation declined
Fixed K32 M1 inner-loop specialization	Exact raw/production/full-output checks pass; 144 operator samples fail advancement	Disabled; model stages skipped
Recurrent prefill fusion	Operator time falls 24–28%; 2B/512 model time falls 2.38%, below gate	Experimental; no advancement
Direct V attention access	Operator gains do not become model gains: 2B/2k is 1.67% slower; 4B/1k effectively unchanged	No adoption
Larger QK grouping	Operator qualification insufficient	No model advancement
Occupied-page gathering	Throughput falls about 2–3%; backing allocation retained; resumed telemetry shows no compacted spans	No speed or full-paged-scheduling claim
OpenCL GPU offload	Zero layers transferred; backend rejects the PowerVR device	No GPU acceleration claim
Vulkan route	Dropped before measured offload	No GPU acceleration claim
Speculative timing gate / hybrid IME-RVV dispatch	Regress tested workloads	Excluded

Long-code MTP outputs differ from direct outputs beginning at generated index 179 for 2B and 303 for 4B. Smaller draft lengths and RS rollback did not fix this. Full-vocabulary target verification does not prove numerical identity between different execution paths. Historical matching 128-token samples therefore do not justify calling MTP universally lossless. Direct decoding is used for the principal results here.

The newer 512-output checkpoint/RS samples match direct outputs on both models. Native RS rollback replay fails strict logit checks. Each model has 18 mismatches, including argmax changes. Full-state restore and same-shape controls pass. This newer evidence narrows the next diagnosis and does not establish general MTP identity or supersede the older long-output traces.

The separate complete-answer MTP pilot completed 24 requests with RS disabled. Total paired task latency falls 8.63% on 2B and 15.22% on 4B; both models pass the predeclared relative fact/code/judge quality gate. The 4B Unicode code differs from direct but passes all 106 functional checks. Absolute usefulness still fails shared interval-code errors and 2B arithmetic, while Chinese prose becomes slower. Checkpoint MTP remains optional. These single-pair task results support workload-specific usefulness, not universal losslessness or general adoption.

The later adaptive infrastructure screen completed 54 timing requests and verified 47 artifacts. Median paired code throughput rises 50.74% on 2B and 53.35% on 4B. The 2B code is capped. The 4B code is shorter under

adaptive MTP and fails reconstruction in every arm. Prose/QA wall time increases 5.77%/4.76% on 2B and 2.97%/3.62% on 4B despite disabled drafting. The relative judge gate passes on 4B, but both models fail absolute correctness and overall timing gates. This establishes bounded throughput, not faster useful answers. Adaptive MTP remains unqualified and optional.

Two complete 4,096-token requests per model and direct/MTP mode demonstrated bounded generation stability, including a roughly 43-minute 4B direct request. These earlier traces are not a long-duration validation of the new routing candidate, and they do not establish indefinite service stability.

7. Full-input cache reuse: separate supporting evidence

Same-process prefix reuse preserves the complete document and sends changed questions through a local server slot. It avoids recomputing an unchanged prefix and is separate from the cold-kernel comparisons. Cached and cold answers need not be token-identical; usefulness is checked on complete answers.

The original alternating quality protocol evaluated both models, 4k/8k documents and six questions: 24 naturally completed cached/cold pairs. An optional MiMo judge scored each pair in both answer orders, supplementing fact, citation, completion, cache and slot checks.

Model / document	Cold / cached mean score	Median cold / cached TTFT	Descriptive decision
2B / 4k	4.50 / 4.50 out of 5	211.11 / 3.90 s	Review required
2B / 8k	4.83 / 4.83	474.31 / 5.01 s	Pilot pass
4B / 4k	4.83 / 4.83	551.91 / 10.80 s	Pilot pass
4B / 8k	4.83 / 5.00	1,271.37 / 14.65 s	Pilot pass

One 2B/4k question scores 2/5 in both arms. The matrix has no universal quality pass. That question does not show a cache-specific regression. One judge and six questions per configuration cannot establish general quality equivalence. These results also do not validate the new GEMM-routing candidate. Length-capped one-, 32- or 64-token speed samples are not complete-answer evidence.

8. Performance evaluator and native operator data

8.1 Purpose and accuracy boundary

The evaluator combines actual instruction traces with measured component costs. Its portable examples retain frozen models, error bounds, and independent test results. QEMU supplies instruction and numerical evidence; it does not simulate K1 cycle timing. A component-cost estimate is not a verified full-model latency prediction. All performance candidates remain available in this evaluator. Its accuracy checks do not use candidate elimination gates.

The V7 audit fixes 28 configuration groups before measurements. It uses 19 groups to calibrate errors and reserves nine groups for independent tests. The groups include 1, 2, and 4 accumulator chains with 0–9 independent integer instructions. Each configuration has two arms and 8, 32, or 128 unrolled iterations. Both arms also change input-register reuse; instruction order is not the only difference.

V7 check	Recorded result
Numerical scope	168 native complete dumps match QEMU for the same ELF; all-one signed 8-bit inputs
Actual traces	336 complete traces; vendor QEMU 10.0.2 executes the IME ELF
Native timing	CPU 0; 512/2048/8192 loops; seven repetitions; raw user-cycle event 0x22
Samples	2,394 calibration samples; 1,134 reserved-test samples
Frozen ratio-error factor	2.1216883936561333

V7 check	Recorded result
Interval coverage	9/9 reserved groups and 27/27 unrolled versions
Point-direction agreement	9/27 versions
Absolute gain error	Median 29.540 percentage points; maximum 50.990 percentage points
Portable replay	Identical key predictions on the local host and K1; 62 tool checks pass

Five two-chain reserved groups contain 15 versions with an incorrect predicted direction. The single-chain measurements are near zero and do not establish a statistically supported change. The reserved set contains no four-chain group. Wide interval coverage does not establish reliable point predictions or an optimization priority order. Nine observed groups cannot confirm a population coverage rate of 95%. The stated coverage requires exchangeable groups within the fixed binary and data scope. The target is the next paired fixture measurement, not a noise-free hardware mean.

The all-one numerical fixture does not establish correctness for arbitrary data. Upstream QEMU 11.1.1 rejects the ELF with an illegal instruction; the failure record remains. Vendor QEMU executes the same ELF with VLEN 256. Frequency snapshots read 1,600,000 kHz but do not establish a constant frequency during each measurement. V7 replay takes approximately 0.44 s locally and 8.11 s on K1 after integrity checks. Those times measure the host tool, not the target algorithm. V7 server replay remains pending.

V5 retains its 7/9 coverage gap. V6's 9/9 aligned-loop group coverage does not transfer to mixed IME loops. Shared-cache traffic, multithread synchronization costs, and full IME operator time remain pending. A gem5 backend calibrated on K1 is not implemented. Independent prediction confidence for arbitrary new algorithms is not established.

8.2 First native operator training batch

The operator report records a successful first batch. The run covers October 12, 2026, 09:24:08–09:25:44 Asia/Singapore. It holds the shared board lock and exits with code 0. The local host and project server verify identical summaries and receipts.

Batch item	Verified scope
Shape groups	16 training groups; eight per operator family
Configurations	32; one-thread and four-thread settings per shape
Numerical dumps	64; paired versions match byte for byte
Timing observations	1,344; all retained
IME worker observations	1,680
File receipts	327 final hashes and 97 hashes fixed before measurements
Independent numerical checks	Reference outputs, guards, unchanged inputs, and state write-back pass
Calibration and reserved tests	Not measured

The IME data call the actual M4 kernel with K32 packing. They exclude production packing, tile scheduling, and backend synchronization. Four threads retain the work per thread and increase total output columns. The SSM data measure the complete convolution graph, including layout conversion and state write-back. They keep total work fixed between one-thread and four-thread settings. The convolution kernel length is 4 and the sequence count is 1. Both families reuse warm inputs from fixed numerical fixtures. These data do not represent the shape frequencies or cache states of full model inference.

The next table first takes the median paired time reduction for each configuration. It then reports the median and range across the eight configurations in that row. Positive values mean a faster candidate. These ranges are observed training results, not prediction confidence intervals.

Operator scope	Threads	Configurations	Median reduction	Configuration range
IME M4 kernel slice	1	8	-12.90%	-14.63% to -10.48%
IME M4 kernel slice	4	8	-12.73%	-19.26% to -1.60%
SSM convolution graph	1	8	37.68%	2.36% to 92.00%
SSM convolution graph	4	8	-0.90%	-41.26% to 82.68%

All 16 IME configurations have negative median reductions. Convolution results depend on the shape and thread count. The batch retains all candidates and all observations. It records 1,853 involuntary IME thread switches. The generic instruction event's user filter still needs an independent check. Foreground-thread cycles do not represent all convolution graph threads.

The first batch chooses the common iteration count from the control arm only. Its shortest formal sample is 0.864482 ms. Fixed overhead and scheduler variation can affect short samples. The revised collector probes both versions twice and raises the common count until all four probes reach 10 ms. This revision has no verified second native batch. Each operator family reserves 19 calibration shapes and nine independent test shapes. Thread configurations of one shape remain in one group. The model must freeze before calibration; its error bounds must freeze before independent tests. Training observations cannot serve as independent test evidence.

8.3 Connection interruption and recovery status

K1 remote access fails during preparation for the second batch. The available records cannot distinguish power loss, reboot, network failure, or remote-service failure. They do not confirm a second-batch start. A client disconnect does not establish that a board task stops. The connection report retains that uncertainty.

The bounded remote client disables connection reuse and limits the total client wait. It records unverified board task status as `unknown` and does not dispatch automatic retries. Six local fake-client tests pass; actual recovery tests remain pending. Board work waits for the requested physical check. After access returns, task, tmux, lock, and batch records must be checked before any authorized restart. This documentation update does not resume the stopped combination or operator collection.

9. Reproduction and deliverables

Use the deployment guide for model preparation and the historical patch stack. The current infrastructure release pins the later `a990751` source and packages its measured RVV/routing code with exact source hashes. The older patch helper targets `5ad05d8`; use separate checkouts. Routing remains opt-in, and unsuccessful SSM/K32 candidates stay disabled. The fresh release verification passes a complete clean build and 104 exact production graph cases per routing arm. All four code responses stop naturally and pass 106 held-out checks each. All 35 archived artifacts and source/build/runtime checks pass; both exit statuses are zero. The one-pair natural-code observations show 508/521 tokens with identical baseline/optimized/prior text. They support package verification, without adding a new statistical speedup or general quality claim.

From the repository root, the latest checks are:

```
python3 spacemit/bench/test-k1-gemm-routing.py
python3 spacemit/bench/test-k1-fast-test.py
python3 spacemit/bench/test-k1-ime-test.py
python3 spacemit/bench/verify-k1-gemm-routing.py \
  spacemit/reports/raw/k1-gemm-routing-20261003-005740
```

To reproduce the board experiment, inspect its protocol and run `python3spacemit/bench/start-k1-gemm-routing.py` with the matching board source/runtime/model prerequisites. The launcher runs the benchmark and collector in tmux under a shared board lock, with explicit time limits. The benchmark guide covers the other campaigns.

The evidence appendix maps every principal claim to source code, provenance, raw records and verification commands. No model weights or private API credentials are included in the report.

10. Release decisions and optional further work

The bounded infrastructure release is verified and reproducible. The current submission report, evidence appendix, source package and test records form the completed deliverable. ARM/x86 comparison is excluded from the release scope.

The strongest replicated long-input result is the activated RVV F16 attention kernel. The opt-in production GEMM routing also improves measured prefill and bounded decode on both required models while preserving tested outputs. Its scope is the current board/runtime and the recorded workload; the release packages the measured mechanisms within that scope. Routing remains opt-in.

Combined mode has independent 8k phase measurements and a complete-answer pilot. The release uses direct decoding and keeps routing opt-in. Baseline citation errors, native RS rollback mismatches and the advanced context cases remain documented limitations; they require separately declared follow-up protocols if pursued. Further unused infrastructure leads are shared K packing across grouped query heads, a Q4_0 RVV lookup-table prototype, and compiler comparisons for measured hotspots. Their platform fit and primary research references are in the October 3 research review. The October 6 source audit identifies channels-major SSM convolution with an RVV channel loop and shared dense FFN activation packing. It also requires separate decode attribution before changes to the worker schedule. The repaired convolution screen now passes 432 exact graph cases in each of four arms. RVV graph time falls 73.72%/78.93% at 32 tokens for the 2B/4B shapes but rises 69.00%/85.10% at one token. This rejects unconditional replacement; full-model state and timing were skipped in that unconditional screen. The subsequent hybrid complete-answer pilot selects RVV only for batches of at least 32 tokens. Three arms pass 432 graph cases each; both models pass 16 exact mixed-chunk/reset/RS-reference-fallback comparisons. All 24 answers stop naturally and all 12 pairs are text-identical. Total answer latency falls 2.92%/2.13% for 2B/4B, while absolute usefulness remains 2/6 and 5/6. Shared model errors and gains below the >3% gate prevent adoption. RS reference fallback does not resolve rollback qualification. No production defaults change. The same-seed clean local reference uses the exact GGUFs and unpatched matching source on generic x86 CPU, without custom K1 paths, flash attention or MTP. All model/source/request hashes pass. It reproduces the 2B wrong sum and missing citations, and both models still fail interval code. Useful answers are 1/6 and 5/6 locally, versus 2/6 and 5/6 on both board arms; one local 4B answer is length-capped. This supports limits of the tested weights/settings without proving causality for every earlier optimization across different CPUs. Local timing is not a matched x86/K1 performance comparison.

The October 7 sustained-decode diagnostic passes 104 production graph cases in each of three arms and eight matched cold requests with exact prompt/token/text identity. Four decode profiles and all 86 artifact hashes are verified. Duplicate FFN branch packing costs only 0.024–0.035% of summed GEMM worker elapsed, so a shared-packing implementation is not pursued. IME assembly accounts for 48.61–66.57% of sampled self CPU; approximately 20% is calling-thread synchronous waiting, which is not proven removable wall time. No new acceleration or useful-answer result is claimed. The fixed K32 M1 inner-loop specialization then passed 480 raw cases, 104 production cases per arm and all actual FFN/full output-head comparisons. Its 144 operator measurements produced no gain above both 3% and control spread; the candidate remains disabled. Full-model and useful-answer stages were skipped, so it adds no measured inference speedup. Separate load scheduling/dataflow, output-head partitioning and dependency- preserving worker-barrier changes remain unmeasured leads.

Remaining advanced coverage limits include unmeasured 4B/32k and 64k workloads, unresolved MTP long-code identity, and unavailable measured GPU offload/full paged scheduling. These are stated explicitly so the submission represents the completed work.

Validation priorities and primary-source practices for these remaining checks are documented in the October 3 validation review.

11. Current handover and remaining work

Area	State at the evidence cutoff	Next evidence needed
Pinned RVV/routing release	Verified; routing stays opt-in	A new protocol for any broader quality claim
Independent B1-r2/C1/C2-r2/C3 candidates	Measured within their stated limits; not production defaults	New release tests before integration
Combined-r1	Failed one declared decode gate; stopped	A separately declared revised-candidate protocol
Dispatch revision	Local path checks and syntax check pass	Native numerical, assembly, operator, and model checks
V7 evaluator	Portable evidence replay passes; point directions unreliable	Target-algorithm costs and independent error data
Operator data	First training batch verified on two hosts	Native checks for longer samples; separate calibration and tests
Board access	Remote failure confirmed; physical cause unknown	Physical check and bounded task/lock inspection
Initial-to-final speedup	No matched complete comparison	Fixed-source, fixed-model direct regression

The strongest replicated long-input result remains the RVV attention comparison. Routing adds measured phase gains on its optimized control. Later individual gains do not override the combined regression or shared quality failures. The project retains negative results, unresolved mechanisms, and incomplete measurements.

Evidence appendix

Evidence cutoff: October 12, 2026, Asia/Singapore. This appendix supports the submission report. File dates identify campaigns; historical work queues do not describe current board activity.

1. Claim-to-artifact map

Submission claim	Authoritative records	Verification / implementation
RVV 2k/8k TTFT improvement on both models	2k requests, 8k requests	Matrix verifier, RVV patch 0009
Actual 16k pairs and 2B 32k feasibility	Resumed measurement matrix, lifecycle raw directory	Same matrix verifier; individual server logs and requests are linked in the matrix report
New routing model results	Routing summary, individual requests/logs	Independent verifier, verification output
Combined routing at 8k, complete-answer pilot and RS replay failures	Staged validation results, summary, 83-artifact receipt	Validation verifier, verification output; 4B/32k explicitly skipped
Native and operator routing gates	Operator records, numerical logs in the same directory	Production graph harness, controller
236 collected artifacts and baseline preservation	Collection receipt, 46-file preservation check	Independent artifact hashing; preservation compares end-of-run board hashes to starting manifests
Source/build/model provenance	Expected baseline, build provenance, tested patch	Generator, launcher
Q8_0 IME1 and M4 scale results	September 24 integrated report, raw arms	Patch 0007, patch 0008
Corrected small-effect statistics	Local verification	Statistics script; fractional Welch degrees of freedom
MTP long-code mismatch and bounded decode traces	Resumed diagnostics, long-code traces	Lifecycle auditor; smaller drafts/RS/target-logit/FA-off evidence in the resumed report
Complete-answer checkpoint MTP usefulness	Results, quality summary, 21-artifact receipt	Protocol, functional checks, scoring recovery; relative gates pass, absolute usefulness fails
Adaptive MTP bounded infrastructure screen	Results, final summary, 47-artifact receipt	Protocol, grading recovery; throughput positive, overall candidate unqualified
Complete-answer cache matrix	Quality report, completed 24-pair summary	Quality runner, document benchmark
Negative compact-layout screen	Fast method results, completed run	Staged test design
Negative IME / recurrent / attention screens	IME scheduling, recurrent fusion, attention infrastructure	Their linked native tests, operator records and gated model runs
GPU and page-gather boundaries	Completed gates, later telemetry and backend findings	Zero transferred OpenCL layers; Vulkan dropped; page gathering retains full backing allocation
Independent bandwidth and routing confirmation	October 6 results, summary, 150-artifact receipt	Protocol, compressed evidence storage; traffic references are not hardware-limit proof

Submission claim	Authoritative records	Verification / implementation
Same-seed local clean-reference quality	Results, summary, verification	Protocol, exact GGUF/source/request hashes, 12 audits and 11 two-order judgment pairs; errors persist, cross-backend causal/speed claims excluded
Hybrid SSM complete useful answers	Measured results, final quality summary, 114-artifact receipt	Completion verification, passing answer; exact graph/state pass, 24 natural answers, adoption gates fail; local scoring files additional to receipt
Sustained decode and negative shared-packing priority	October 7 results, summary, 86-artifact receipt	Independent verifier, verification; eight exact matched requests, four decode profiles, native identity; packing declined, CPU shares are not wall-time savings
Unused research opportunities	October 3 GitHub/arXiv/Scholar review, October 6 source audit	Dated primary-source/source reviews; convolution and decode packing subsequently measured; M1 kernel scheduling and worker-barrier implementation remain unqualified

2. Baseline identities

Source state	Role
Official SpaceMiT fork `5ad05d8`	Clean starting point for the packaged patch helper
`6562c22` plus separately selected options	Earlier recurrent/MTP integration state
`a990751` plus recorded wide-RVV source changes	Later board baseline; build-specific provenance is required
Original library versus generated mode 0	Routing operator control for unintended effects of the isolated library build
Generated mode 0 versus mode 1 or 2	Routing model comparisons; the baseline already has wide attention/layout 0

Historical documentation values with different quantization, model derivative, prompt length or runtime state are not matched baseline arms. In particular, the old Q4_1 documentation rates must not be used to calculate a controlled Q4_0 optimization percentage. Baselines from different campaigns must not be treated as one continuous cumulative curve.

3. Latest model identity

The routing run hashes the exact local files; model preparation is described in the model tools guide.

Model file	SHA-256
`Qwen3.5-2B-MTP-Q4_0-embQ4_0-dv64k.gguf`	`56f0ddd90dfa0e2456af6cb4718ece419678d21a75d65e27353319b5e431a2fa`
`Qwen3.5-4B-MTP-Q4_0-embQ4_0-dv64k.gguf`	`0adf6cce5df53921606033c39a13fae2054543b78af0e7ff98860e86f078ac0f`

Model weights remain outside this repository. The current kernel comparisons do not replace the model, alter its Q4_0 weights or trim input tokens.

4. Re-run local verification

From the repository root:

```
python3 spacemit/bench/verify-results-matrix.py \
  spacemit/reports/2026-09-27-resumed-measures.md \
  spacemit/reports/raw/2026-09-25-lifecycle
python3 spacemit/bench/verify-k1-gemm-routing.py \
  spacemit/reports/raw/k1-gemm-routing-20261003-005740
python3 spacemit/bench/test-k1-gemm-routing.py
python3 spacemit/bench/test-k1-fast-test.py
python3 spacemit/bench/test-k1-ime-test.py
```

The matrix verifier checks 21 rows / 105 numeric cells. The routing verifier checks archived hashes, numerical logs, operator and model gates, input/output counts, actual runtime configuration and per-phase kernel markers. The three local regression scripts contain 2, 6 and 3 checks respectively. These local tests supplement the board runs; they do not substitute for them.

The routing run's environment telemetry allowlist omitted `SPINE_K1_GEMM_ROUTE`. Actual mandatory kernel markers record mode, eligibility and bypass for each phase and establish activation independently. The verifier requires those markers and checks the environment value if recorded. Future runs also capture the environment field. Original raw records were preserved.

The 236-artifact receipt covers collected board evidence. The independently generated local verification and preservation records are additional files, not members of that original collection count. Native binary output dumps remain on the board; numerical logs and summaries record their identical SHA-256 values. End-of-run board hashing independently confirmed those dumps.

Fixed K32 M1 screen: negative evidence

The October 7 M1 specialization results have the following evidence: the frozen summary, 144 operator samples, 211-artifact receipt, independent verifier, verification output and 40-file remote preservation check. Both exit statuses are zero. All 480 raw cases, 104 production cases per arm and six actual FFN/full-248320-column shapes per arm pass exact outputs and guards. Recreated route-3 baseline hash is exact; generated assembly removes the inner counter/branches while retaining dot/load/FMA order and fallbacks. No operator clears both 3% and control spread. Full-model state, cold ABBA and useful-answer stages were skipped. Candidate disabled; no new inference or quality gain enters the submission. Independent analysis files are outside the original 211-artifact receipt.

5. Submission document roles

The October 6 repaired SSM screen is supported by its summary, operator samples and 97-artifact receipt. All four arms pass 432 exact output/history cases. Warm graph gains at 32 tokens accompany single-token regressions, so model-state and inference timing stages were skipped. These measurements support no full-model gain.

Material	Use
SUBMISSION-REPORT.md	Current report; edit this file before a new export
SUBMISSION-REPORT.tex	Standalone TeX export of the report and this appendix
SUBMISSION-REPORT.pdf	PDF export for review
REPORT-BUILD.md	Export commands and document update rules
This appendix	Evidence and reproduction references accompanying the report
Current documentation guide	Operational status and navigation
Integration guide / benchmark guide	Deployment and executable procedures
Dated reports and experiments	Campaign-specific details and historical decisions

Material	Use
`raw/`	Reproducibility records; preserve original measurements
September 24 LaTeX and PDF	Historical export; predates later findings and is not the current submission
Root FINAL-REPORT.md	Historical short-prompt campaign, with later annotations; not the current submission

No required final-report template, author list or external submission destination has been supplied. The report is organized for review; it has not been submitted externally.

Current release package

The release manifest pins the exact source revision, patch and measured files, model hashes and matched route0/3 profiles. Offline clean-source verification checks the 3,266-file snapshot and exact patch round trip. The fresh full board build and complete-answer verification passes in `k1-release-check-20261007-125712`. It checks 104 exact production graph cases per routing arm. All four Unicode code answers stop naturally and pass 106 held-out checks each. All 35 archived artifacts, 18 staged code files and three collector files pass checksum verification; post-run source, binary, compiler and vendor-runtime checks pass. Board/collector exit statuses are zero. The independent result records exact paired/prior text identity and the descriptive natural-answer timings. ARM/x86 comparison is excluded by user instruction. Release smoke timings do not add a new statistical speedup claim.

6. Evidence added after October 7

Claim	Records	Tools and limits
Independent C1, C2-r2, C3, and B1-r2 results	Candidate review, progress snapshot, phase comparison CSV	Version comparison; C0 already has RVV attention and route 3; no cumulative speedup
Combined-r1 fails one decode gate	Stop audit, saved original summary, independent audit	Audit tool; 96 first-round requests and 48 comparisons; no second round or complete-answer stage
Dispatch revision remains local	Protocol, manifest, local checks	Full numerical, target assembly, operator, and model checks remain pending
V7 wide intervals and incorrect point directions	V7 report, evaluation, calibration, cross-host check	Evaluator guide; 9/9 group coverage does not establish reliable point predictions
First native operator training batch	Operator report, plan, observations, summary	Operator tools, numerical receipts, transfer receipt; training only
Access failure and unknown second-batch state	Connection report	Bounded client; no automatic retry or new board task

The October 11 progress report and comparison CSV retain their original snapshots. The October 12 stop audit supplies the later combination decision. A saved summary is not a new independent comparison of the 536 formal state dumps. The report retains that evidence gap.

7. Portable evaluator replay

Run these commands from the repository root. The output directory must not exist.

```
python3 spacemit/bench/qemu-screen/predictor/model-bundle.py \
  --include-ime-audit --output lab/k1-portable-v7
```

Check the generated archive against its external SHA-256 receipt. Extract the archive. Run this command from its `k1-models-v7/` directory.

```
python3 run-examples.py --output ../k1-replay
```

The replay checks fixed evidence and frozen model identities. It does not collect new native measurements or qualify a new algorithm. Complete archives and binary dumps remain outside Git. Full archive checks require those retained files. The portable examples include the data needed for their own replay.